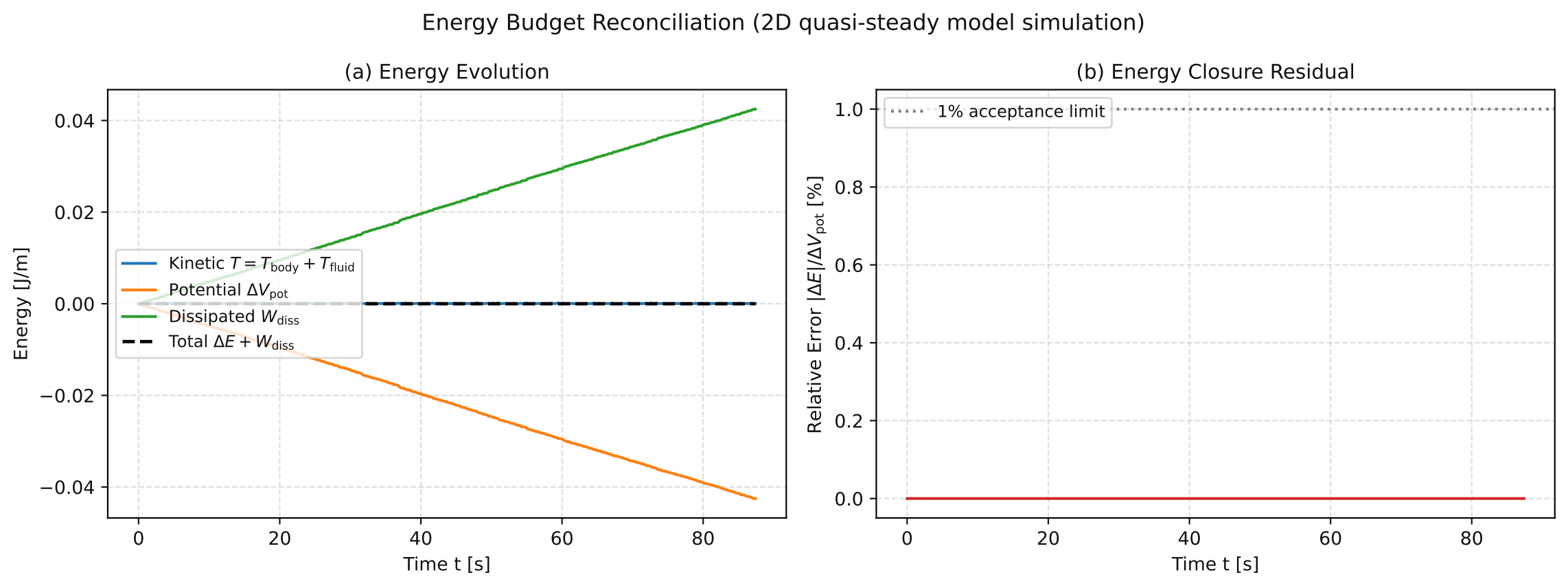
Energy Budget Reconciliation (2D quasi-steady model simulation)
(a) Energy Evolution
Energy [J/m]
Time t [s]
Kinetic $T = T_{body} + T_{fluid}$
Potential ΔV_{pot}
Dissipated W_{diss}
Total $\Delta E + W_{diss}$
0.04
0.02
0.00
−0.02
−0.04
0
20
40
60
80
(b) Energy Closure Residual
Relative Error $|\Delta E|/\Delta V_{pot}$ [%]
Time t [s]
1% acceptance limit
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80